15 Two solid cones are mathematically similar.

Cone **A** has a volume of 120 cm^3
Cone **B** has a volume of 960 cm^3

Work out the ratio of the surface area of cone **A** to the surface area of cone **B**.

Volume ratio = 120 : 960 = 1 : 8.

1 mark
Volume factor8

Cube root gives length ratio 1 : 2.

1 mark
Cube root, then square

Area ratio = $1^2 : 2^2 = 1 : 4$.

1 mark
1:4

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(Total for Question 15 is 3 marks)